



UIL Computer Science Competition

Invitational B 2025

JUDGES PACKET - CONFIDENTIAL

I. Instructions

1. The attached printouts of the judge test data are provided for the reference of the contest director and programming judges. Additional copies may be made if needed for this purpose.
2. This packet must remain CONFIDENTIAL. Additional copies may be made and returned to schools when other confidential contest material is returned.

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Problem 10	Ruth
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Problem #1
60 Points

1. Akash

Program Name: Akash.java

Input File: NONE

Test Input File: NONE

Test Output To Screen:

```
2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20
2 3 5 7 9 11 13 15 17 19
2 3 5 7 11 13 17 19
```

Problem #2
60 Points

2. Amy

Program Name: Amy.java

Input File: amy.dat

Test Input File: (indented lines are continuation of previous line ...)

```

100
cdiy4ug08c34nilpulxiebcodc8rlnlg0 2494357888
n6jrwuxerp05tlc8n11sczf5r310v4fmz0n8ioyft7wn8hnb6w4de41uy0aemha49rp9sttm8gb6gk6j
    c2tjzbbnum3cb9k 308915776
ak4dr6haa5h27w67w7uup7mkwxmik0bv2tgvedtx59kpdusna68yrn01cmhybkf 81
kcqe4czvu027tnth166f87qznd906w 161051
067auwbe06193as 3404825447
9exioke31xitpxl9nlvwojx9zifp7x7kk5r5i8hwmvst4uil2glrr8hlqkpyesluyf0c7b4mjb3f6
    1564031349
sdlfbu2 15752961
zhq2pvzkdccjcbodc4jhj5q55sbe9hy33lewrdbxbrghd 75
jrm2p094oft59znp5x4rnpe8zht3ptj1o8gbnyifibxy4sv 10648
83robmwg5omgznbqtt968g4el7ib2jesd9hqcqppq4otf7 78364164096
028693y9go7ess19htmbvwlili1doo4smu5eeig2xb98v 3404825447
kislrwytmqzznpea0j97u98uljz0e688kybnzyv9vir7nr5bschhspgoc0cuf9jfgk9ukmvbbyneqmgf
    w5p7vpg7ikqzb 1522435234375
95a2v1mkin9wisi8k8gcppzlns8omzeczfc8u4 10030613004288
2t652jjylyqi86mfy7ci9bq6xuqr5smifwxuwcjmf5h2iawwevu5e 3938980639167
fz09jv2b86kgjxe 42618442977
i0i 7056
tf4ia2nud63r1j4vlp1dvbujiya5xcsb1b2gal25l0nrx2cna4pn2vflmtt9ubh02mkvnfukv
    941480149401
3k81tflhxkkgxb529y015wybbarq25nz54prjt 20151121
p4qmjmc12zra7iqva9ps0uj33sfkottis7k18xe5p3i09cfu4k821m2apmec660us4xybw00
    68574961
0ld 82
bdauayxmwwr2q3bnloy7mb1jlrcqomt305pxwjth1 12151280273024
01l6i27ur4iy54knq0zpgtd5lhsf4v4wy2hbe0gof99dlfm7ulysb3wyhhcsqwdtirkzukbulsvqs0lq
    537824
v9fb7juwgs2 456976
178y0fksjp6r4k8gb39ab6l5kvgyd7dx154zp5t5twluhhcl7t38u4zyrel89r5 29509034655744
wvker518l0hu5bi1kmfw0f16dnt 98
frd8bpgqe82ft0kgc4ioydw6nt0ielqwnrmfcgthuetmleswbp8j12k0b5vok6ok3jyzdgtj9vu
    20151121
6njcjrnsbpp2259dlitwx82rxs17i8p33zfey8ukbk299vzsq7ii0cyntaoeltgib153g2yqrtxotwn9
    ru4327mwbws 36
xxhi112xqhlhbr7ndf93fxb38gpae8hcncwmgp0j1ezf18ikhiqav 117649
11z49jykcjcvg0gmej0b0du26ntvkkfkdrezydl2a2ykt5801wmf4okuyk41stib18pujqljg2bpz6
    smlcqpzgzk4m2 1
ohimppne53fnr5hys3ays5jxgykicvxfbg3c3ivdaoomqcdcnli2ul6ajdomhflh2tsy9u014mtndqti
    yejykyk0e 319277809664
4zn28j60f81quq6 421875
pbjoqoabr1s4xhfkhnmk9z0ftcn8lutjv 636056
4ujant72ud2awo7n8wsm0q953z4e2a9li0iqw8koquqb4xv3gwfe3vwo8dvm4gtujnhf7nhg16ni47s2
    w93kqnl18jfr7 17210368
14q9mdg27sitz9mlkjegefherb39jqrzv3ikc6jhi9kft09d841azmq7mzdhcqpwm 5929
m76hvazqnvqeptz16ublych0uccpz5ux118189gtshb 4084101

```

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k9g1c7mg45qbezzbzfr5cgys468oykdb2dz1f6n8dgra0etklshcl9kkmihkwrcz
1677721600000000

~ *input is 101 lines and continues... - view problem's dat file for complete input* ~

~ *output is on next page...* ~

~ *Amy continued...* ~

Test Output To Screen: (indented lines are continuation of previous line ...)

914163252529637798881025019167268717918746
22850593529215792259964458342380955946880742147246985619882505768676299608710531
090054965077765294708132093455884447375489644797821026497938
13758545660314233470730635760296446545577511960705803244018284028310890683329414
43258628677042176
303467087945366731590134505676925966716511
64929589831256
43756956320830719176779678580917768360934877254174072245607833706428916845902165
8885952107678710482884423619729
4974
111159419365842289282913515472213465521852013668836412372956563374
1315071886628323834684049120985471159649354281018005772944795814811129
4963608617944918181428679924706605506341271472266715087241216
3173347864831048567494800476888925474693803217634051970433178
35776161709256228553452869291371670231952958280231475067109385272198255293609737
15302594055733481785608793238511084657787586524150415
13745721932068853668872024076999107337217114112
7738355805796841535396820356931710391017583602935359851872417534792702
5187282882202
6
43278419120650650950700561594481861763604102937763014427012378199044041461231085
9493733871452761212068
6842200846550254755097578173203539708151101911218618
59417857342345801554875791202546051743846591729103142803793548255772685704186811
20212598947284559955526605
568
19058248882050184005295495841204706825518339259712293
59369020392785801165247207629587354673239482162567461747300004210878513864222863
384540994863691372431364951223141884388
288027607231
37766135405196124548985710624547260422305241522072249674988876810310907169959106
68515
10689107506064634924423560430527255471584
26205252382443564202808336950882907329838313089762088547038424799810561463804961
902172083068540403399329928474
11674144619527907862805338279133588163327013171583248172377967639883685693028297
4584939114053628368372944886748404030500135265750611474251776
259086211510674193532007761260413141420504865794236946227452555167789429852939
15129691426908168590195718409757130259671809070371889631401846061289256658164673
5062081091813502365411336573225931623528175304412792470630301696
10156728252682276880203325465322582798644166245084345948014496719088569262366274
929323356323614960072214873848326211981127825627
524027068967664255
3584983586141845996927695564675481275164879548
31647719059156321889633380457132391901683050968659590935560846706265272171630975
3645879001848541829832349699677167765791778011885656886749

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```
24360214019854055874910089721249448010938520755389650891281217476066184185751199
350230697251000078
2041319852872926427681027883594458098401313049104988081335829
23913335290835085392386039422168992833437934444795919811461756364437673826568467
90636
57442018917231789259452427571469248871047606885033226860639612351593812219201350
0099408367591598506522786115124248148199986373068918
```

~ output is 100 lines and continues... - view problem's out file for complete input ~

Problem #3
60 Points

3. Dante

Program Name: Dante.java

Input File: dante.dat

Test Input File:

```
15
4 15
10 14
8 11
1 100000
100 1000
1000 10000
10000 100000
19 475
537 5384
4858 37574
123 234
345 456
567 789
890 12345
473 56375
```

Test Output To Screen:

```
4
2
0
9592
143
1061
8363
83
610
3330
21
19
35
1320
5623
```

Problem #4
60 Points

4. Girish

Program Name: Girish.java

Input File: girish.dat

Test Input File: (indented lines are continuation of previous line ...)

```

57 19 26 99 36 31 10 69 140 171 71 48 40 56 119 83 20 93 114 142 20 150 124 126
    177 111 38 189 175 117 83 81 23 21 11 88 198 13 83 206 180 65 70 35 121 78
    136 117 48 135 126 153 130 155 167 198 61 132 78 50 110 32 10 33 83 202
    157 133 114 122 7 147 59 75 30 178 85 14 99 66 191 151 74 74 145 135 30
    165 91 71 59 33 72 125 137 183 7 44 24 209 41 130 155 81 70 153 188 96 78
    205 63 55 126 55 110 63 192 97 90 175 39 115 195 22 26 19 196 170 156 134
    152 4 79 126 58 159 5 12 140 82 66 42 206 186 142 88 168 68 181 64 148 107
    1 74 193
78 4 154 131 156 200 206 154 168 141 110 44 72 142 71 32 173 151 109 102 53 147
    42 39 44 153 93 210 58 3 89 102 196 63 147 69 169 154 141 142 142 38 139
    207 155 111 137 107 1 179 198 82 62 17 154 183 28 68 199 88 208 11 66 171
    209 26 55 146 91 149 154 78 63 99 163 108 203 115 104 191 143 25 83 88 208
    80 50 10 200 136 1 94 139 140 139 161 56 186 185 157 85 32 44 185 157 172
    126 22 26 28 105 31 121 65 103 116 71 164 114 183 210 59 174 209 115 82 74
    60 97 62 94 136 94 134 113 130
149 122 168 202 199 177 68 3 155 186 187 14 148 171 178 168 50 31 47 213 211 21
    26 199 194 18 155 54 202 5 111 163 65 22 97 78 185 60 193 144 173 24 173
    140 185 36 27 90 114 141 179 44 131 124 29 154 160 13 180 154 76 129 77
    197 53 23 52 56 86 114 57 152 191 156 112 166 149 133 4 37 29 152 127 9 39
    193 174 144 49 7 193 93 8 182 77 72 120 167 38 123 16 165 54 13 61 136 137
    183 173 29 132 92 210 30 5 67 7 60 27 202 77 109 156 82 167 50 6 74 208
    144 12 121 55 104 32 5 118 135 39 204 199 6 188 176 93 100 94 28 180 206
    100 150 39 122 23 206 213 121 59 203 79 192 120 81 47 107 177 21 24 66 164
    186 199 125 167 129 179 69 54 127 165 98 8 12 89 181 21 49 189 179 167 72
158 145 174 75 72 42 15 212 30 18 110 44 206 178 66 93 15 92 160 181 199 69 59
    94 156 139 71 193 35 125 129 36 81 65 88 154 156 164 65 199 168 110 101 44
    65 169 130 92 153 57 149 131 194 194 52 88 176 82 63 104 3 192 171 21 202
    27 153 179 107 151 81 139 161 37 201 71 15 139 111 88 24 82 16 11 50 115
    36 160 152 140 209 3 166 62 49 173 177 165 183 147 117 191 186 92 39 7 167
    214 157 181 156 200 61 4 146
49 78 112 171 188 199 75 4 106 206 191 39 36 47 70 211 48 85 99 193 206 86 193
    209 137 165 60 92 39 107 99 22 30 106 56 159 207 63 193 134 172 85 69 91
    135 73 4 154 182 20 50 25 21 131 182 204 17 99 90 53 104 7 128 83 179 175
    56 174 48 66 100 67 59 41 118 86 107 191 115 198 125 130 60 174 54 208 150
    86 152 126 170 22 183 144 215 50 160 6 8 150 46 15 29 176 177 35 88 19 191
    127 164 148 90 108 112 14 138 204 32 139 46 152 172 139 63 136 208 207 11
    185 147 52 174 115 113 89 121 151 8 61 75 108 136 98 117 138 172 205 29
    140

```

~ input is 31 long lines and continues... - view problem’s dat file for complete input ~

Test Output To Screen:

Median Mean: 119.89

Mean Sum: 16442.67

Mode Kinda: 156

Problem #5
60 Points

5. Howard

Program Name: Howard.java

Input File: howard.dat

Test Input File:

```
15
12 Armstrong PG 18.5 7.8 9.9 3.4 2.7
15 Devens PG 17.5 7.8 9.9 3.4 2.7
13 Harden SG 35.6 7.6 24.3 10.5 4.5
34 O'Neal C 28.5 1.5 17.3 12.6 1.8
9 Caruso PG 8.5 3.4 5.4 0.8 0.5
77 Doncic SG 35.6 7.6 22.3 10.5 4.5
78 Valanciunas C 15.6 0.9 10.5 10.5 3.6
23 James SF 28.5 10.1 19.4 7.3 2.4
35 Durant PF 31.4 3.4 19.6 9.3 2.3
36 O'Neal2 C 28.5 2.0 17.3 12.6 1.8
24 James2 SF 28.5 10.1 19.4 10.3 2.4
37 Durant2 PF 31.4 3.4 19.6 9.3 1.3
57 ABCD PF 1.0 1.0 1.0 1.0 1.0
59 ABCDE SF 1.0 1.0 1.0 1.0 1.0
63 ABCDEF SG 1.0 1.0 1.0 1.0 1.0
```

Test Output To Screen:

```
PG: 12. Armstrong
SG: 77. Doncic
SF: 23. James
PF: 37. Durant2
C: 36. O'Neal2
```

Problem #6
60 Points

6. Hulin

Program Name: Hulin.java

Input File: hulin.dat

Test Input File:

```
100
3230 18908 34117 38084 45759 46652
663 25170 30761 55088 62164 83934
16936 21451 36301 48373 62509 72058
20129 42169 54953 58085 62110 63439
25906 27899 39155 65813 66882 81428
20372 47750 51770 61639 71650 76047
14387 33287 48269 57144 80357 82017
21911 26674 28256 30107 40990 68385
6973 51880 65408 82897 84009 86115
2590 20076 28784 46035 51780 52771
25108 47394 49731 71012 83090 83746
1457 5501 27367 36977 56759 58834
5140 9157 9641 12159 18509 48322
265 2243 34340 39704 79963 82518
18206 36800 39026 63654 65247 66620
16175 19652 30954 44794 49734 57757
4754 10168 20337 20619 31877 44309
37457 40104 41227 48271 73429 86312
1554 22897 24542 44319 45379 47600
7413 9506 26742 32772 44763 56572
39240 46214 52179 55021 56382 83322
5411 62917 73857 75872 85210 86054
21199 37821 39929 48537 61965 81353
7590 23662 35662 68113 75250 83075
8899 16040 25319 27642 28407 85612
2469 12715 14475 26566 63750 79255
12080 13955 25963 44652 52324 60082
3831 14977 19214 20233 59588 69651
3591 8677 37304 47886 56624 64408
7270 13711 39620 63841 80318 83402
34599 37708 57180 64970 76094 83249
12361 27230 50160 50867 62526 68476
1270 8311 31817 53416 54063 82154
3608 14054 24823 29143 48784 64651
8139 22295 27218 28136 47778 62720
385 15119 34535 62658 67506 76061
4260 31832 32540 39838 64108 71461
13801 39251 46921 55777 58461 62551
26592 43973 49604 53338 57542 72054
26180 34263 43946 51340 55060 55678
11229 25958 39850 52392 60032 81138
5708 25707 35667 52912 55109 61322
8586 32340 38904 45035 51914 52144
6379 19484 46876 51489 63519 64319
5856 8093 24192 58942 76166 78478
12358 14539 29631 47949 58313 65061
6874 11121 31925 72425 75304 79100
8091 14251 51610 53447 56245 66750
14770 19082 55334 74389 80239 82187
```

~ input continues on next page... ~

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~ *Hulin input continued...* ~

```
23497 36901 66862 69558 69822 70135
10760 36687 48990 63647 74943 84574
18198 32157 51832 57186 57591 83629
5365 5483 10874 13204 50638 55854
12591 19835 48462 61761 62238 85088
1023 8737 16224 50007 78026 85246
6226 39993 58207 62830 72960 80483
605 37461 42611 49261 59896 84864
15324 20494 30961 65501 68070 69797
9528 35994 39998 45867 64494 77736
10103 49240 51119 60594 62780 68058
12817 25707 28186 40463 52036 67159
17669 23630 55368 56209 76104 79135
12858 29329 34074 54323 82771 85235
26014 33965 35694 40624 63956 66821
6277 24040 32477 59797 83542 84306
34904 42258 67437 67476 67885 76398
19703 24802 26914 35160 58262 59392
14879 45334 63821 74024 74117 75965
2520 40815 58875 61624 65674 68705
2672 6567 12438 25615 50407 69342
3822 65138 65497 67709 69622 86376
39110 45390 49830 58212 67547 82651
555 6799 26766 33302 62080 72721
7245 17250 17552 18096 32267 35692
15289 16275 27013 28153 41426 72545
25165 27509 44471 61313 76030 80020
6696 31188 43278 47463 56239 67014
11870 25346 29783 61535 74978 76274
4720 13733 37454 52723 54255 64411
22321 29898 31560 59483 72499 85730
11429 14119 29345 40540 50307 51423
7919 20862 34862 38456 57089 61209
4143 16810 30600 41989 45226 59052
863 16277 41331 59141 70973 76225
5863 11325 13886 23719 70312 71702
7177 11344 21862 45611 64919 83127
4924 55322 68925 69849 78036 85634
9362 20340 32227 45533 49758 85865
19942 33488 35579 39980 63530 82750
3793 4932 17935 19492 32655 49531
5514 20713 24587 39599 55806 82387
3449 15345 34919 41166 51324 56822
4476 43249 55423 64450 66028 80093
12113 24051 30127 43450 70515 81820
8551 14867 26030 44362 45765 77502
3239 3779 28455 28804 45217 47048
320 1864 19761 27284 63877 71085
6185 44368 47715 62055 70011 70048
504 27955 49876 51165 65071 79130
5754 11319 28062 51026 53225 58188
```

Test Output To Screen:

62

Problem #7
60 Points

7. Julia

Program Name: Julia.java

Input File: julia.dat

Test Input File:

100
8183003
4439451
1737777
5499056
8060342
4666622
7085826
8093669
3220227
3820519
512576
1883291
201041
6949530
516274
7404092
4823407
2898642
8761818
1881682
3914971
7722895
8783656
9323789
4173253

~ Input continued ~

1060195
1424072
431593
7754275
6704050
3044771
3340773
9028192
3235899
2559418
2228695
4609166
508360
1157087
7162069
9245231
3014847
4221295
5209802
3444925
1737677
8396602
9808658
4820971
1935891
8014509

~ Input continued ~

2644705
4047313
5515685
4162637
9901473
8054316
3120468
6565963
2626077
7627975
2884214
3146137
9177488
3932460
9910913
4634665
6246520
8641390
2974570
1938035
3774976
5341689
2678329
2087585
8251802
5664932

~ Input continued ~

8367963
4554237
8326992
2204591
9464080
3467915
6459352
1693601
559076
2853802
402153
6145202
8618957
1382302
8731079
6700992
4936848
326867
5331758
4783688
4104091
3433611
8789577

~ Output on next page ~

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~ *Julia continued* ~

Test Output To Screen:

551255
311958
130739
380732
543570
326754
482111
545635
231586
271364
42478
140871
18065
473440
42761
502222
336975
210007
587605
140765
277585
522356
588995
622647
294550

~ *Output continued* ~

82846
108775
36282
524301
457758
219795
239608
604186
232642
187090
164629
323048
42167
89818
486931
617754
217820
297717
362055
246579
130733
564665
652813
336809
144522

~ *Output continued* ~

540684
192889
286311
381801
293862
658539
543174
224874
448972
191628
516356
209052
226575
613494
278752
659109
324701
428598
580048
215092
144664
268337
370598
195161
154965

~ *Output continued* ~

555598
391335
562865
319456
560298
162980
631351
248066
442174
127681
45999
207040
34015
422133
578645
105859
585680
457559
344380
28152
369950
334395
290030
245801
589370

Problem #8
60 Points

8. Nicolas

Program Name: Nicolas.java

Input File: nicolas.dat

Test Input File:

1000

37 72

```

...#.....#...##.##.....#.#.##.##.....#...#...##.##.###.....#.....###...#
##.....#.#.#.....#.....#.#.#.....#.....#.....#.....#.....#.....#.....#.....#
#.....#.#.##.##.....#.....#.....#.....#.....#.....#.....#.....#.....#.....#
.#.##.##.....#.....#.....#.....#.....#.....#.....#.....#.....#.....#.....#
...###.....###.##.....#.#.....#.#.....#.....#.....#.....#.....#.....#.....#
.##.##.....###.##.S.##.##.##.....#.....#.#.....#.....#.....#.....#.....#.....#
.##.##.##.....#.....#.....#.....#.....#.....#.....#.....#.....#.....#.....#
.....#.....#.....#.....#.....#.....#.....#.....#.....#.....#.....#.....#.....#
.#.....#.....#.....#.....#.....#.....#.....#.....#.....#.....#.....#.....#
.##.....#.....#.....#.....#.....#.....#.....#.....#.....#.....#.....#.....#
##.....#.....#.....#.....#.....#.....#.....#.....#.....#.....#.....#.....#
#.#.#.##.##.....#.....#.....#.....#.....#.....#.....#.....#.....#.....#.....#
#.#.#.##.##.....#.....#.....#.....#.....#.....#.....#.....#.....#.....#.....#
.....#.....#.....#.....#.....#.....#.....#.....#.....#.....#.....#.....#.....#
.#.###.##.##.....#.....#.....#.....#.....#.....#.....#.....#.....#.....#.....#
#.....#.....#.....#.....#.....#.....#.....#.....#.....#.....#.....#.....#
###.#####.....#.....#.....#.....#.....#.....#.....#.....#.....#.....#.....#
..#.#.#.....#.....#.....#.....#.....#.....#.....#.....#.....#.....#.....#.....#
..#.....#.....#.....#.....#.....#.....#.....#.....#.....#.....#.....#.....#
###.....#.....#.....#.....#.....#.....#.....#.....#.....#.....#.....#.....#
#.....#.....#.....#.....#.....#.....#.....#.....#.....#.....#.....#.....#
.....#.....#.....#.....#.....#.....#.....#.....#.....#.....#.....#.....#.....#
###.....#.....#.....#.....#.....#.....#.....#.....#.....#.....#.....#.....#
.#.....#.....#.....#.....#.....#.....#.....#.....#.....#.....#.....#.....#
.....#.....#.....#.....#.....#.....#.....#.....#.....#.....#.....#.....#.....#
.##.....#.....#.....#.....#.....#.....#.....#.....#.....#.....#.....#.....#
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..##.##.##.....#.....#.....#.....#.....#.....#.....#.....#.....#.....#.....#
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```

~ Input file contains 100846 lines of data, some lines very long – see problem’s dat file for complete input ~

Problem #9
60 Points

9. Pranav

Program Name: Pranav.java

Input File: pranav.dat

Test Input File: (indented lines are continuation of previous line ...)

```

15
10 7 3 5
10 10 10 10 10 10
8 15 3 7
10 10 10 12 10 10
10 10 -10 10 10 -10
10 -10 -10 10 10 -10
5 9 1 59 3 4 29 5 18 4 9 29 3 8
10 7 3 5 10 10 10 10 10 10 8 15 3 7 10 10 10 12 10 10 10 10 -10 10 10 -10 10 -10
    -10 10 10 -10 5 9 1 59 3 4 29 5 18 4 9 29 3 8
10 -7 3 5 10 -10 10 10 -10 10 8 -15 3 -7 10 10 10 -12 10 10 -10 10 -10 -10 10 -
    10 10 -10 -10 10 10 -10 5 9 1 -59 3 -4 29 5 -18 4 9 -29 3 -8
34 34
34 35
1 1 1 1 1 9 1 1
6 8 4 9 5 1
1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1
212 17 94 78 212 93
    
```

Test Output To Screen:

```

Winner 15
Tie 30
Winner 22
Winner 32
Tie 10
Winner 10
Winner 118
Winner 221
Winner 96
Tie 34
Winner 35
Winner 12
Winner 18
Tie 11
Winner 518
    
```

Problem #10
60 Points

10. Ruth

Program Name: Ruth.java

Input File: ruth.dat

Test Input File: (indented lines are continuation of previous line ...)

```

4
3 0 2 0 4
2 1 5 3 1 0 4
2 2
13 14 9 10 6 9 14 7 6 14 2 14 11 19 11 3 8 12 17 6 4 11 9 10 4 3 9 7 12 1 14 11
    5 17 0 9 15 15 16 14 6 14 5 6 7 11 17 11 6 16 18 7 18 6 10 6 16 15 16 4 1
    2 17 12 3 13 16 6 9 17 2 7 15 7 18 8 11 10 14 7 16 14 5 15 7 10 10 9 3 7
    19 15 5 13 2 16 2 0 1 9 4 14 12 5 7 13 14 14 8 19 3 2 6 12 1 19 13 9 0 16
    10 18 8 14 9 19 4 15 5 4 17 19 18 8 19 7 19 19 3 2 14 12 12 0 9 15 15 13
    16 7 16 2 18 9 6 19 10 17 17 2 3 14 19 9 8 17 3 9 1 12 0 14 17 5 19 14 2
    12 7 9 0 14 8 1 11 2 0 12 6 11 3 12 5 15 12 4 16 15 1 8 16 17 11 18 1 2 17
    8 3 10 18 6 13 17 2 7 8 1 10 1 2 19 19 2 11 18 16 6 13 18 10 3 8 8 9 10 18
    9 4 9 5 3 9 10 4 18 15 4 5 17 8 11 19 14 7 8 10 3 13 2 3 6 7 11 9 7 17 14
    1 18 19 13 14 1 0 17 18 10 9 1 9 0 0 1 19 0 17 15 5 15 5 7 10 6 5 2 13 6 7
    12 7 9 0 9 0 7 15 3 18 10 18 9 13 18 16 1 0 9 6 19 19 19 8 18 13 19 0 2 0
    5 17 11 1 18 15 18 5 16 19 12 7 5 7 8 1 18 19 14 10 4 7 13 11 4 18 2 5 17
    15 13 15 12 1 8 5 18 10 2 19 18 7 5 10 9 14 10 14 8 5 12 17 4 6 17 14 14 0
    10 18 18 4 15 2 19 11 0 18 11 1 17 2 17 13 8 17 5 15 2 1 19 3 5 7 19 15 11
    16 16 1 7 9 13 0 12 16 8 3 6 8 1 1 8 8 15 5 17 19 13 17 15 0 10 6 4 6 5 16
    17 12 3 8 16 0 11 9 2 6 0 3 4 8 16 3 15 1 3 11 15 1 16 18 16 16 18 0 3 11
    11 13 5 13 11 14 5 10 14 12 16 15 10 8 2 16 10 7 0 2 19 2 6 3 1 3 11 8 2 5
    10 0 16 2 1 10 14 5 16 4 19 8 14 14 1 0 7 19 17 3 18 19 3 3 4 12 18 15 4 7
    14 15 1 10 16 7 4 15 11 16 0 1 16 2 2 15 19 15 4 6 18 4 17 9 9 11 13 8 6 5
    11 6 17 4 1 1 9 10 6 5 8 6 1 8 0 1 11 10 9 13 18 18 15 19 17 8 10 14 12 3
    4 12 8 7 4 4 5 4 4 11 15 5 19 4 17 17 5 13 2 16 2 9 18 19 17 16 3 9 17 0
    18 1 9 19 18 6 11 17 12 10 4 8 8 16 14 18 16 11 0 12 3 7 4 19 16 6 10 3 13
    1 18 6 6 8 16 7 5 6 2 2 17 16 5 10 4 6 12 10 9 15 15 5 19 5 6 0 14 15 12
    10 18 17 4 16 4 0 0 1 13 16 9 17 10 17 4 4 7 17 0 11 9 7 8 12 0 3 12 2 17
    4 11 7 4 4 18 1 14 1 16 11 11 7 13 17 3 5 12 1 7 16 17 8 2 17 0 17 0 7 12
    3 11 3 16 3 0 19 15 6 6 4 7 11 0 8 11 10 17 14 10 9 14 13 18 12 13 13 13 1
    15 8 3 11 1 8 16 16 10 9 14 0 4 19 8 6 8 10 7 17 18 18 13 14 11 9 9 10 3
    10 14 3 16 15 12 3 4 6 8 18 11 18 15 14 13 4 14 8 8 11 12 2 16 1 16 17 16
    1 8 7 12 2 2 4 10 15 4 1 13 15 4 16 16 1 17 2 14 0 9 6 6 9 19 5 10 15 7 11
    5 15 3 6 2 3 3 16 11 0 2 8 6 0 7 7 4 12 5 7 16 15 1 17 3 19 17 15 17 2 5 5
    3 9 3 2 9 10 10 15 4 12 18 19 14 0 12 10 5 12 2 10 13 8 6 1 15 18 0 1 7 4
    0 9 0 8 11 10 0 15 3 6 7 3 4 17 0 18 8 9 7 14 19 10 15 3 13 6 7 5 0 10 3
    12 5 13 11 3 1 6 12 13 0 19 13 11 16 16 2 19 12 0 14 10 2 4 15 5 12 14 4 4
    14 6 4 11 2 17 19 14 1 13 16 7 13 6 10 5 14 19 18 4 18 16 13 12 14 11 18 1
    18 8 10 8 11 6 16 10 5 5 8 17 1 8 14 9 19 8 1 19 14 15 1 5 4 8 8 9 6 7 14
    5 2 0 11 17 17 0 5 8 0 2 19 3 7 12 9 11 13 5 14 19 2 8 11 3 10 5 18 15 13
    19 16 13 6 0 10 7 1 1 0 7 10 14 4 6 17 18 7 16 10 6 6 11 11 6 4 17 14 17

```

~ 5th line of input is extremely long and continues... view problem’s dat file to see complete data ~

Test Output To Screen:

```

7
9
0
3986153

```

Problem #11
60 Points

11. Sarac

Program Name: Sarac.java

Input File: sarac.dat

Test Input File: (*indented lines are continuation of previous line*)

```
20
556 164
R72 P230 R96 P255 R248 P52 R193 P126 P138 P186 R284 R136 P194 R164 P115 R76 R252 R82 P122
P34 R79 R268 P38 R1 P93 R183 P152 R10 P226 R13 P14 P200 R98 R143 R196 R181 R274
R43 R174 P55 P125 P235 R77 P62 R251 P147 R23 P88 P242 P176 P185 P159 P249 R151
P170 R247 P94 P86 P120 R53 R122 R269 R114 P219 R213 P9 R4 P199 R58 R156 P202 R244
R187 P212 P259 P10 P21 P145 P68 R56 P84 P248 P66 P27 P223 P105 R46 P83 R153 R209
P141 P80 R204 R145 R166 R65 R231 R5 R276 P40 P7 P20 P90 R250 R249 R291 P100 R198
R220 R67 R282 R162 P101 P148 R235 R134 P78 P1 R188 R202 R78 R185 P239 P103 R292
R91 R37 R115 R266 R157 R99 P179 P160 P207 R12 P253 R52 R120 R51 P133 P77 P6 R59
R221 R11 P30 P91 R222 R8 R146 P184 P216 P2 P132 P157 R3 P197 R271 R100 P227 P99
R81 P192 P8 R179 P204 R152 P67 P106 R177 R110 R55 R278 R295 R216 R6 P169 P236 R95
R69 R168 P45 R160 R259 R182 P173 P195 R117 R2 P165 P156 R24 R40 R139 P193 R234 P17
P217 P183 P102 P211 P172 R130 R101 P129 R199 P3 P29 P75 P196 R211 P108 R230 P252
R283 P39 R210 R201 R225 P56 P225 R255 P233 R165 R272 P81 R57 R228 P70 P82 R80 P190
P140 P151 R192 R227 R270 R20 R17 R35 P51 P154 R34 R90 P89 R232 R219 R86 R14 P143
P95 R116 R124 R155 P119 P50 R135 R92 P72 P87 P137 R63 P58 P59 P234 R19 R93 P13
R214 P43 R41 P4 P49 R175 R74 P231 R28 R112 P131 R50 R286 P92 R205 P42 R127 R89
R159 P244 R73 P139 R103 P116 R60 R262 P16 R264 P28 P5 R224 P163 P258 R71 R47 P205
P48 R195 R238 P57 R22 P224 P167 P26 R137 P189 P214 P36 P124 P63 R260 P155 P73 R64
R167 P229 R229 R180 R111 P104 R104 P127 R25 P246 R254 P118 R191 R189 R194 R18 R178
R49 P175 P37 R70 R42 R267 P162 R61 R277 R27 R223 R138 P149 R273 R190 P98 P203 P15
R281 P158 P201 R39 P109 R94 P136 P251 R236 R125 R263 R31 R296 P111 P232 P22 P64
P113 R126 R141 P18 R261 R83 R29 P161 R163 P96 P142 P107 P46 R154 R208 R88 P112
P250 R66 R253 R171 R132 R215 P254 P11 P76 R280 R203 R242 P134 R15 R257 P71 R287
P128 R105 R226 P123 P47 R147 R258 R87 R186 R48 P243 P41 P146 P245 R9 P61 P31 P256
R176 R158 R290 R218 R26 P54 R108 P210 R118 R140 R21 R38 P187 R97 R45 P191 R84 R150
P257 P238 P206 R245 R121 P60 R133 R85 P177 P24 P198 P213 P208 R172 P182 P153 R161
R288 R246 R243 R279 P237 P150 P114 P79 R149 P19 R200 R75 R109 R102 P164 R7 R285
P221 R275 P144 P240 R148 P12 P117 P215 R36 R54 R169 P174 P33 R173 P23 R237 R16 R30
R184 R119 P178 P168 R233 R123 R131 R240 R142 R207 R197 P44 R44 R32 P85 P222 R62
P171 P121 P188 R33 R217 P110 R128 P228 P209 P32 R241 R129 R212 P247 R107 P65 P180
P166 P53 R294 P35 R265 R206 P130 R144 P181 R239 R106 R170 P135 R256 R289 P69 R293
P97 P25 P241 P260 R68 P218 P220 P74 R113
R49->P258 P235->R3 R129->P245 R210->P164 P51->R97 P3->R99 R195->P163 R164->P105 R80->P149
R62->P208 R238->P7 R201->P251 R290->P208 P174->R225 R227->P122 R252->P182 R231-
->P58 R23->P4 P101->R176 R204->P116 R290->P35 R162->P152 R294->P240 R90->P169 R136-
->P122 R38->P147 P85->R261 P180->R77 P245->R52 R147->P6 P95->R289 P54->R217 R176-
->P122 P135->R267 P29->R69 P50->R20 R245->P215 P199->R2 P89->R210 R68->P183 R21-
->P45 R78->P189 R124->P122 R249->P167 R64->P131 R290->P182 R129->P42 P42->R125
R198->P86 R57->P34 R118->P137 P155->R51 P253->R177 R111->P258 R57->P207 P149->R229
P195->R295 R260->P29 R73->P170 R125->P61 P140->R236 R91->P58 R20->P238 P91->R150
R66->P33 P136->R235 R271->P143 P88->R252 R67->P4 P8->R105 R41->P38 P6->R124 P23-
->R144 R101->P223 R14->P72 R180->P183 R188->P236 P100->R175 R119->P203 R268->P130
R269->P221 R133->P217 P207->R270 R54->P39 P23->R284 R128->P236 R282->P45 R50->P223
P138->R185 R228->P257 P86->R279 P85->R87 R100->P15 R22->P120 R243->P10 P10->R290
R256->P110 P69->R267 R33->P82 P40->R173 R246->P113 R173->P79 P10->R174 P143->R96
R19->P37 R181->P203 R88->P233 R18->P229 R49->P106 P183->R101 R259->P195 R135->P4
P133->R285 P68->R80 P133->R167 P182->R245 P219->R192 P58->R286 R126->P187 P65-
->R221 R286->P126 R191->P75 R62->P239 P40->R163 P215->R240 P161->R130 P109->R278
P80->R286 P117->R16 R132->P186 P211->R145 R183->P115 R14->P14 R247->P50 R166->P31
R190->P62 R98->P107 R206->P130 R294->P237 R230->P224 P179->R227 R227->P224 P57-
->R223 R78->P21 P96->R141 R284->P74 R1->P156 R63->P130 P64->R223 R252->P120 R162-
```

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>P72 P102->R148 R296->P201 R196->P101 R277->P19 P227->R184 R61->P49 P28->R123
P135->R204 R136->P140 R113->P14 R18->P245 R17->P259 P121->R7

~ Above data is the line with number of test cases and is followed 3 lines for the first test case ~

~ Input file contains 62 lines of data, most very long – see problem’s .dat file for complete input ~

~ Sarac continued ~

Deadlock free; all is well
Deadlock exists; not good...
Deadlock exists; not good...
Deadlock exists; not good...
Deadlock free; all is well
Deadlock exists; not good...
Deadlock exists; not good...
Deadlock exists; not good...
Deadlock exists; not good...
Deadlock exists; not good...
Deadlock free; all is well
Deadlock exists; not good...
Deadlock exists; not good...
Deadlock exists; not good...
Deadlock exists; not good...
Deadlock exists; not good...
Deadlock free; all is well
Deadlock exists; not good...
Deadlock exists; not good...
Deadlock free; all is well

Problem #12
60 Points

12. Varsha

Program Name: Varsha.java

Input File: varsha.dat

Test Input File:

```
10
2 5 6
###ooV
#oooVV
**oV#o
***Voo
%%VV##
####VV
###V%
#oooo%
V*oo%
VVoo%
3 3 3
VoV
###
#V#
**V
*VV
###
###
VV#
###
1 8 10
##oo@@%%V*
**#@V%*##o
oo@@@VVVo#
#oo@#@Vo##
ooo**VV##
#####V##
#####
##VVVV####
4 2 2
#V
oo
#V
oV
##
```

~ Input continued ~

```
VV
V%
V%
3 1 3
Vo#
Vo#
Vo#
2 4 1
o
V
o
o
o
V
V
o
1 3 3
#o%
@*o
*%@
1 3 3
VoV
oVo
VoV
4 4 4
VVoo
oVV#
###V
##VV
%%%%
%V%
%VV%
%%%%
VVVV
VooV
VoVo
VVVV
VVVV
```

~ Input continued ~

```
####
#VV#
####
4 4 4
VVoo
oVV#
###V
##VV
%%%%
%V%
%VV%
%%%%
VVVV
VooV
VoVo
VVoV
VVVV
####
#VV#
####
```

Test Output To Screen:

```
5
6
6
7
3
3
0
1
25
13
```